

YOLO (You only look once): Assisting the Blind and Visually Impaired (BVI) Using Context-Aware Temporal and Spatial Visual Interfaces

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Research Question:

How can the context and memory-based disadvantages of LLMs be supplemented or overcome, allowing for long term use of visual aids for BVI individuals?

Summary

Researchers have long been interested in developing aids for the blind or visually impaired (BVI) to verbally describe scenes [1]. BVI users can use mobile phones or augmented reality glasses furnished with these aids to understand their surroundings and navigate. These aids use machine learning models to detect objects in the user's scene via the device's camera.

Vision-enabled Large Language Models (LLM) like GPT-4V can accurately describe images using natural human language. As a result, they have been in these assistive aids to provide verbal descriptions of the objects detected. If prompted by the user, the aid will only provide descriptions for relevant objects. For example, if the user prompts the aid with “find a silver laptop on the desk, and monitors or other office objects might be around it.” The aid searches for objects relevant to the prompt (e.g., laptops, desks, monitors) as the user scans the environment using their camera. Once the aid detects the objects in the camera's field of view it will describe the scene with “a silver laptop with an Apple logo and black keyboard is opened on the white desk.” These aids tend to be verbose and repetitive because they need a more sophisticated long-term memory of visual descriptions. In other words, previously navigated spaces or paths should not be reintroduced by the aid upon revisiting; instead, only visual changes or new elements since the last visit should be described. This capability is not only more user friendly, but it is especially important for individuals who have cognitive disabilities that impair their memory. To achieve this a better temporal spatial representation of the scene needs to be used instead of simply storing previous descriptions.

[1] Chang et al. 2024. WorldScribe: Towards Context-Aware Live Visual Descriptions. In Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology (UIST '24)

Methodology

- (1) Design a spatial-temporal representation of AR scenes. This representation should describe objects in the scene, their relationships, and the time of detecting them.
- (2) Detect objects using a machine learning model (e.g., LLM) and store the current spatial-temporal representation of the scene in a database.
- (3) Create an end-to-end AR system that can generate natural language and relevant descriptions of the user's scene using the stored spatial-temporal representation database.
- (4) Conduct a user study to evaluate the usability of the assistive aid with BVI users

What will ACM research members learn?

(1) Assistive Technologies for BVI Users:

- Gain an in-depth understanding of the challenges faced by blind and visually impaired individuals in navigation and scene comprehension.
- Explore state-of-the-art assistive technologies for BVI users and their applications in real-world scenarios.

(2) Machine Learning and Vision-Enabled LLMs:

- Learn about Vision-enabled Large Language Models (LLMs), such as GPT-4V, and their capabilities in image understanding and natural language generation.
- Develop skills in fine-tuning and applying LLMs for context-aware and task-specific applications.

(3) Spatial-Temporal Scene Representation:

- Explore methods for constructing and maintaining spatial-temporal representations of dynamic environments.

(4) System Design and Development:

- Gain experience designing and implementing augmented reality assistive systems that integrate machine learning, computer vision, and human-computer interaction principles.

(5) Human-Centered Design and Usability Testing:

- Understand the principles of human-centered design, with a focus on creating accessible and intuitive interfaces for individuals with disabilities.
- Learn to design and conduct user studies to evaluate the system's usability and effectiveness.

(6) Research and Communication Skills:

- Learn to conduct a literature review, synthesize research findings, and identify gaps in existing technologies.
- Develop skills in documenting research progress and presenting findings in academic papers in journals or conferences.

Resources Needed

- Lab Computer: Necessary for using the Magic Leap 2 (AR goggles) and running 3d environments and image detection models fast enough. Also for training an LSTM capable of creating a spatial-temporal representation of a scene.
- Magic Leap 2: Enterprise Grade AR goggles that allow for object scene understanding and direct camera access for image detection capabilities and developer tools.

*As an aside, these requirements have already been satisfied.

Bib

Chang et al. 2024. WorldScribe: Towards Context-Aware Live Visual Descriptions. In Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology (UIST '24)